E Caption reranking

Same five candidates $y_1 \dots y_5$

D · Decoder likelihood

Sum token log-probabilities
Raw sequence log-probability

Min–max normalization
over the same 5 candidates

No length normalization.

D

C · Image–text alignment

CLIP image / text encoders
Image I + candidate caption

Cosine-based CLIPScore
2.5 max(cosine, 0)

Min–max normalization
over the same 5 candidates

C

O · Object consistency

Mentioned object extraction
from candidate caption

Supported object labels
from YOLO detections

Detector-based
object consistency

Normalized object score
(consistency + 1) / 2

O

$S(y) = 0.3\ D(y) + 0.4\ C(y) + 0.3\ O(y)$

Select highest-scoring candidate

Final caption

Equal scores: earlier beam wins.

References and CHAIR are not used for selection.